Quotient moments: three-support control verified

November 13, 2026. Unsigned producer evidence [pcc:unverified].

Role. This checks the earlier explicit prediction using a supplied, known norm witness A, B. It tests certificate construction and recovery, not finding that witness from public data.

General pairing foundation. For any field, monic F, and nonzero P with $\deg P < \deg F$, Lean constructs a polynomial Q below F with

$$[X^{\deg F-1}]((PQ) \bmod F) \neq 0.$$

A complementary monomial suffices. If $\gcd(F, U) = 1$, the same detection is possible for $PUQ \bmod F$. No characteristic or separability assumption is needed. These are polynomial detection lemmas; the complete general Hankel-radical theorem remains a separate obligation.

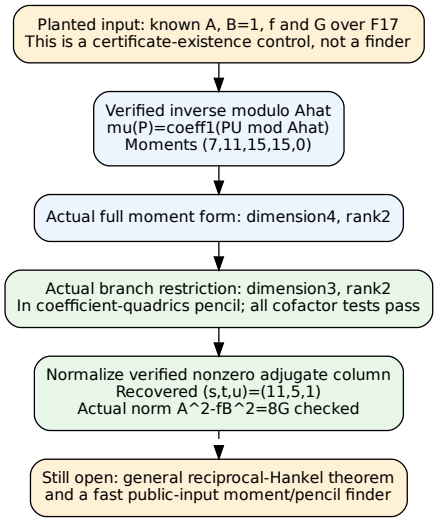
Exact control. Over $\mathbb{F}_{17}$ take

$$f = X^3 + 2X + 2, \quad A = 5X^2 + 11X + 6, \quad B = 1, \qquad A^2 - fB^2 = 8X(X-5)(X-6)(X-3).$$

For $\widehat{A} = (X-16)(X-9)$ and $U = 7X + 6$, the inverse identity is checked. For every polynomial P over $\mathbb{F}_{17}$,

$$\mu(P) = [X]((PU) \bmod \widehat{A}) = 12(P(16) + P(9)).$$

The first five moments are exactly $(7, 11, 15, 15, 0)$ and $\mu(f) = 0$.



Actual coefficient map, rank certificate and evidence

The full four-dimensional moment form has rank2. Restrict by the actual branch $A(0) - 6B(0) = 0$, writing $A = tX^2 + sX + 6u$, $B = u$. Lean checks the coefficient equations

$$2ts - u^2 - 3t^2, \quad s^2 + 12tu - 12t^2, \quad 12su - 2u^2 - 12t^2$$

and their complete polynomial expansion. Their matrix pencil at parameters $(15, 15, 11)$ equals the actual restricted moment matrix

$$S = \begin{pmatrix} 15 & 15 & 15 \\ 15 & 0 & 5 \\ 15 & 5 & 14 \end{pmatrix}, \qquad \operatorname{adj} S = \begin{pmatrix} 9 & 1 & 7 \\ 1 & 2 & 14 \\ 7 & 14 & 13 \end{pmatrix}.$$

Its rank is2, so its corank is1. Every required cofactor pairing vanishes; the preceding generic recovery theorem is instantiated with all conditions proved. The normalized third adjugate column is $(11, 5, 1)$ and the actual recovered norm equals $8G$. No numerical-rank assumption is used.

Formal map. Module `NormQuotientMoment` contains the general pairing detectors, the quotient/evaluation identity, actual branch restriction, pencil membership, full/restricted ranks, coefficient expansion, stationarity and recovered norm. Exact declaration names and hypotheses are in the README. Module `NormQuotientInteger` separately proves the universal integer lifts used by the native certificates, rather than attaching a stronger integer claim to a finite-field proof.

Evidence. Core source `3d8417de4`; integer source `cbb5e6372`; Lean4.35.0 with unchanged pinned dependencies. Core build005 passed3149 jobs; expanded build passed3151. Actual tracked-source gate passed90 declarations with zero additional trusted axioms. Separate import-inclusive fresh receipts are retained per module. Earlier theorem declaration hashes are unchanged. Failed builds and visibility/normalization errors are retained.

Native records `IDC1h91f69b513509f342` and `IDC1he7eb817a6f348de6` each have32 statement-derived and32 extra-stream points with exact universal proof-source hashes. The native checker does not execute Lean or establish a search bound. All standing is unsigned producer evidence; claim bindings and full logs are in the versioned evidence directory.

Limits and next test. This is $a = 2$ with constant B; it does not establish nontrivial lower-block reciprocity, general degree/chart cases or a public-input finder. Next: a generic CRT coefficient-reciprocity lemma and its Hankel-radical consequence. The $2^{o(n)}$ search bound, $n = \lceil \log_2 r \rceil$, remains absent. No performance claim, paid job or scaling campaign was run. Sources: pinned monic-division, quotient-coordinate and matrix declarations; KB unavailable, no novelty asserted for these classical ingredients.